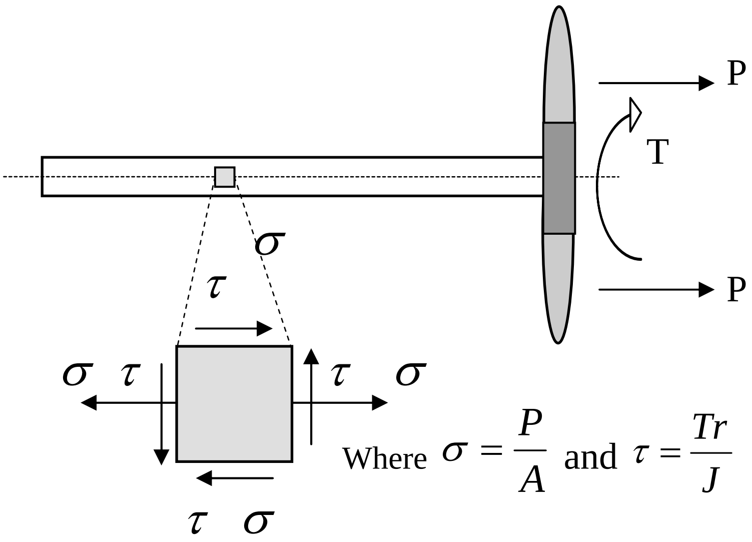
So far, we have been concerned with problems that have involved only one type of stress.

The majority of engineering components and structures are subjected to loading conditions that, at any point in the material, a “COMPLEX” state of stress exists, involving normal (Tension/Compression) and shear components in various directions.

A simple example of a complex stress situation is given by the prop shaft of an airscrew. Torque is passed through the shaft, thus producing torsional shear stress.

This force, P, puts the shaft in a direct tension and that generates an axial tensile stress.

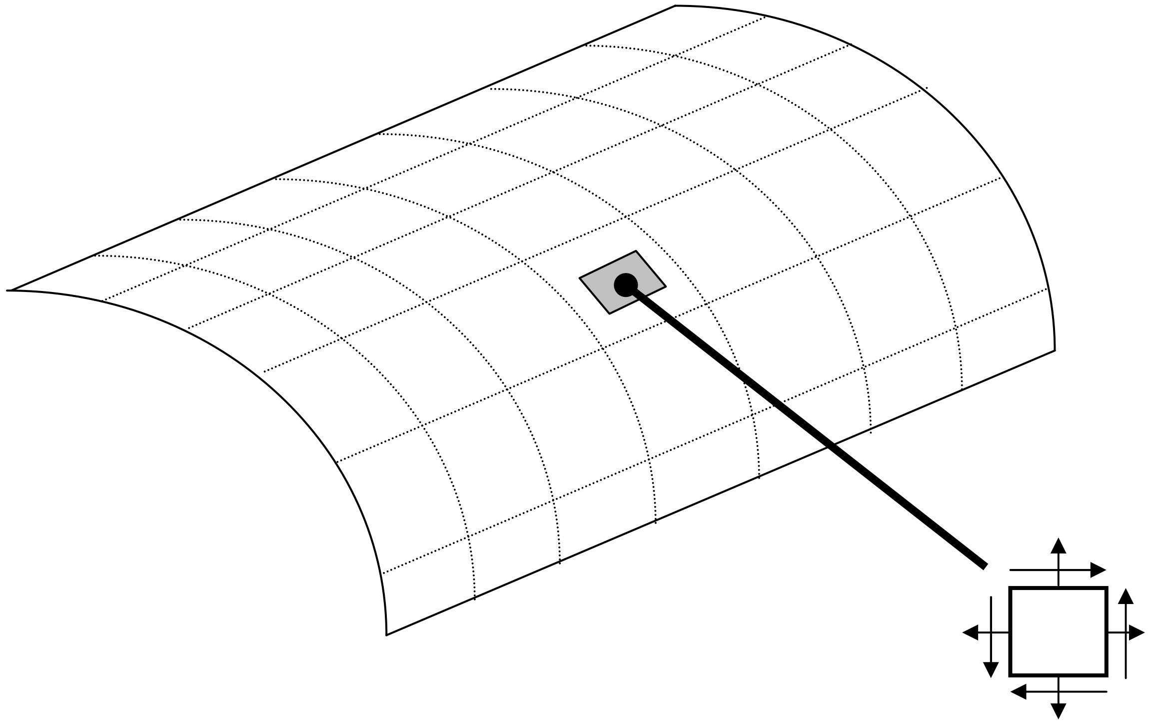


Where $\sigma = \dfrac{P}{A}$ and $\tau = \dfrac{Tr}{J}$

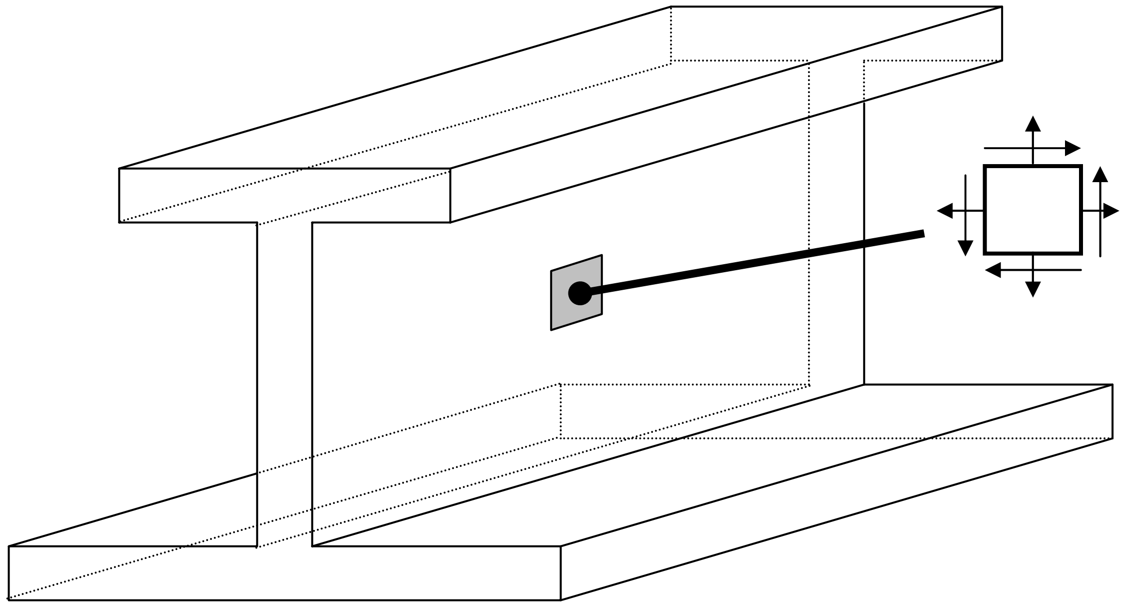
It is not necessarily sufficient to be able to determine the individual values of applied stress in order to select a suitable material from which to make this shaft because, as will be seen later, on certain planes within the element more severe conditions of stress and strain may exist.

The identification of these maximum stresses is an essential step in the **Design Process**.

Another example is the state of stress in the skin of an aircraft.



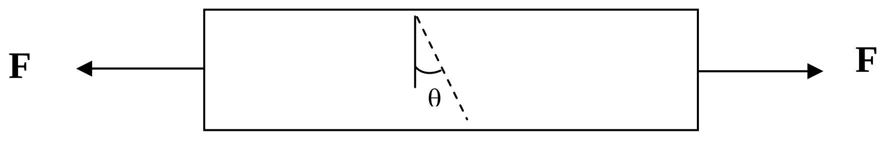
A further example is the state of stress in the web of a beam.

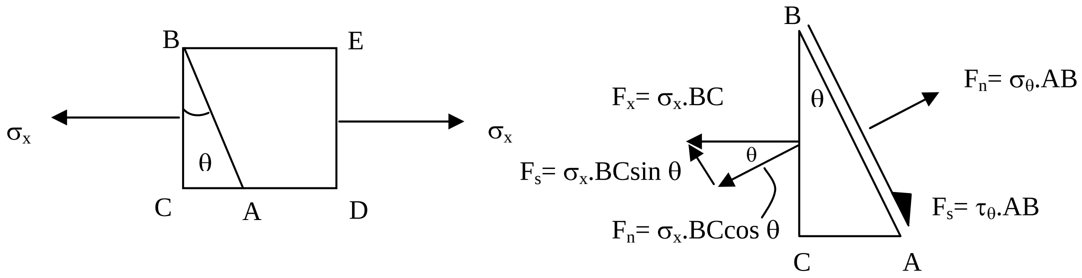


So far, we have only considered stress set up on the plane perpendicular (Normal Stress) or parallel (Shear Stress) to the direction of loading. But we need to be able to calculate direct and shear stresses on any inclined plane within the stress element.

Firstly consider uni-axial loading in a simple bar

The bar shown below is subjected to an axial tensile force, F. The bar has unit depth and unit width.





Where AB=(1/cosθ) and BC=1. Hence BC/AB= cosθ.

$$\sigma_\theta AB = \sigma_x BC\cos\theta$$

$$\therefore \sigma_\theta = \sigma_x\left(\frac{BC}{AB}\right)\cos\theta$$

$$\boxed{\therefore \sigma_\theta = \sigma_x\cos^2\theta}$$

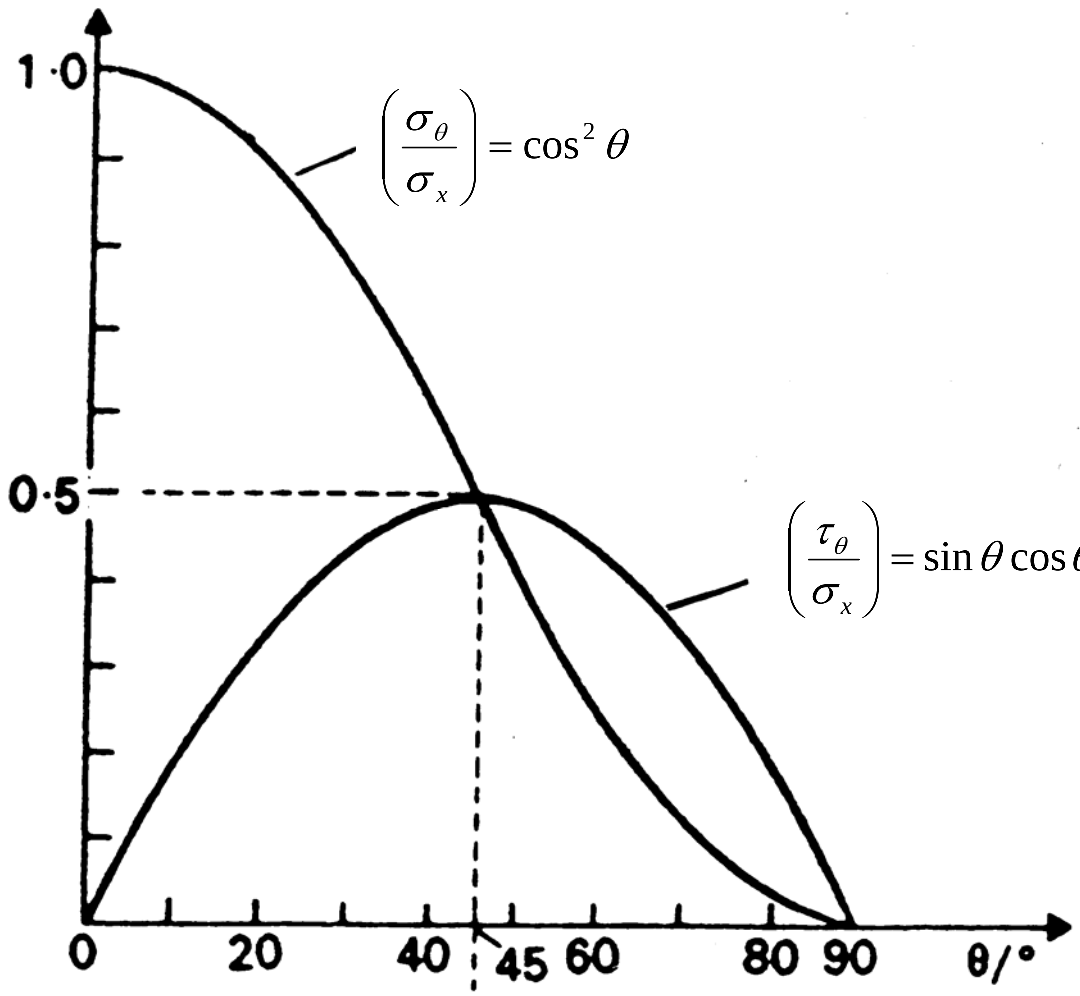
Similarly,

$$\tau_\theta AB = \sigma_x BC\sin\theta$$

$$\therefore \tau_\theta = \sigma_x\left(\frac{BC}{AB}\right)\sin\theta$$

$$\boxed{\therefore \tau_\theta = \sigma_x\cos\theta\sin\theta}$$

The plot below shows a graphical representation of the relationship between σ_θ and τ_θ in dimensionless form, by dividing through by σ_x.



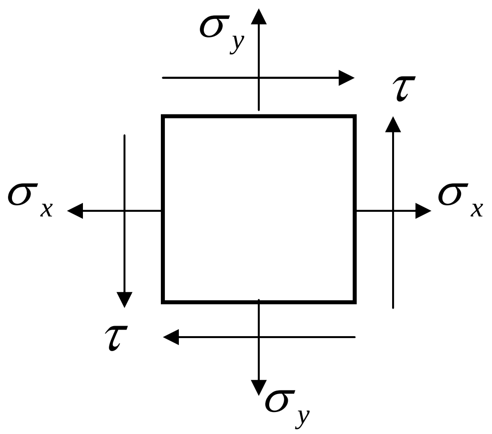
This plot shows that:

σ_θ has the maximum value σ_x when $\theta = 0°$
σ_θ has the minimum value zero when $\theta = 90°$ (indicating no transverse stress in the bar)
τ_θ has the maximum value ½ σ_x when $\theta = 45°$

In general, for materials whose shear strength is less than half the tensile strength, direct tensile loading will result in failure along planes of maximum shear stress.

A 2 Dimensional Element

To enable us to analyse a 2-Dimensional element of material under a 2D stress system we need to apply a similar method of considering force equilibrium for the element shown below:

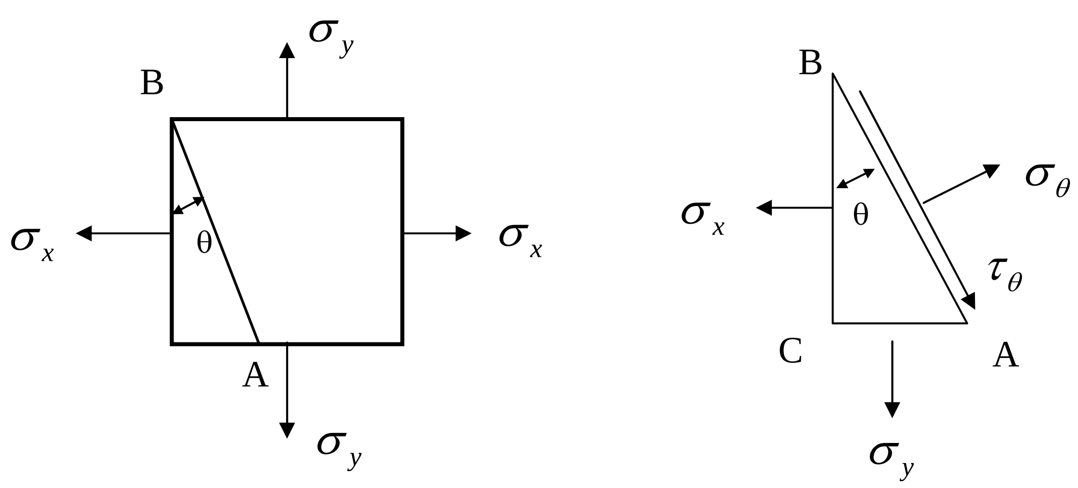


To simplify this process we can break the analysis down into the following two cases and then superimpose the results.

Case 1 - Element subjected to direct stresses only
Case 2 - Element subjected to shear stresses only

Case 1 Consider an element subjected to direct stresses only

Consider an element of unit thickness and unit length sides subjected to tensile stresses in the X and Y directions (bi-axial system).



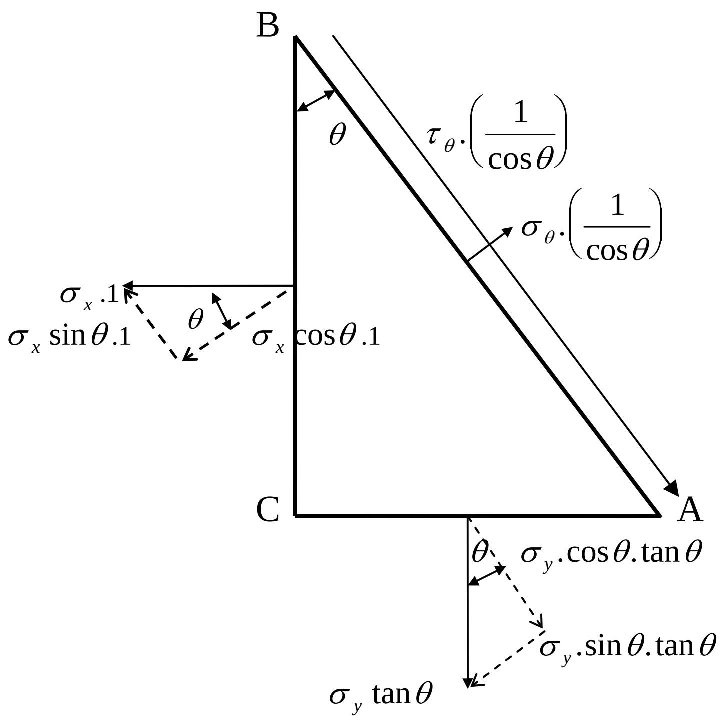
Consider a corner cut off the element by a plane AB shown above.

Length of side BC = 1, Area of face BC = 1

Length of side AB = $(1/\cos\theta)$, Area of face AB = $(1/\cos\theta)$

Length of side CA = $\tan\theta$, Area of face CA = $\tan\theta$

Considering the equilibrium of forces on the triangular element ABC, the forces on AB, BC and CA must be in balance:

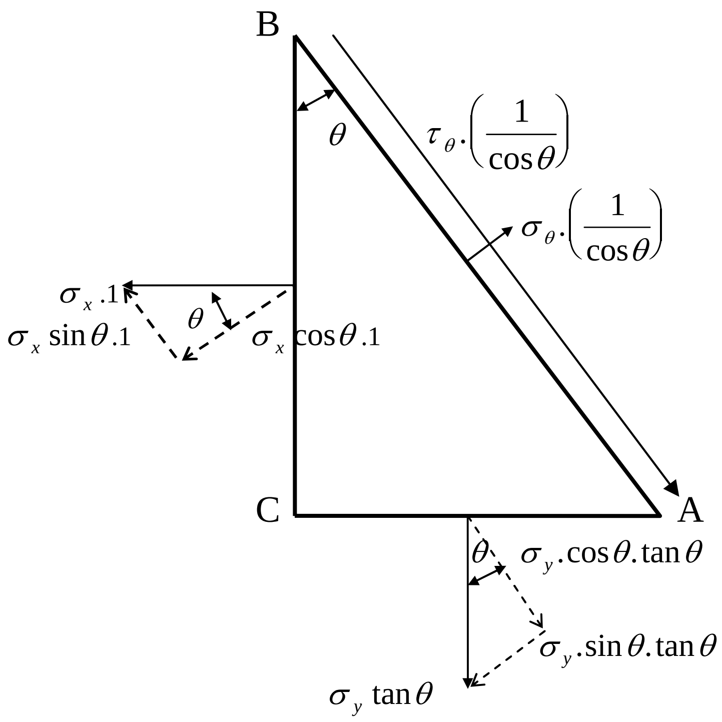


Resolving forces in the direction of σ_θ (perpendicular to plane AB):

$$\sigma_\theta.\left(\frac{1}{\cos\theta}\right) = \sigma_x \cos\theta.1 + \sigma_y \sin\theta \tan\theta$$

$$\boxed{\sigma_\theta = \sigma_x \cos^2\theta + \sigma_y \sin^2\theta}$$

Considering again the equilibrium of forces on the triangular element ABC, but this time resolving parallel to plane AB:



$$\tau_\theta.\left(\frac{1}{\cos\theta}\right) = \sigma_x \sin\theta.1 - \sigma_y \cos\theta \tan\theta$$

$$\tau_\theta = \sigma_x \sin\theta.\cos\theta - \sigma_y \sin\theta.\cos\theta$$

$$\boxed{\tau_\theta = (\sigma_x - \sigma_y)\sin\theta.\cos\theta}$$

In terms of double angles where: $\cos^2\theta$ = ½ (cos 2θ + 1), $\sin^2\theta$ = ½ (1 - cos 2θ), $\sin\theta\cos\theta$ = ½ (sin 2θ)

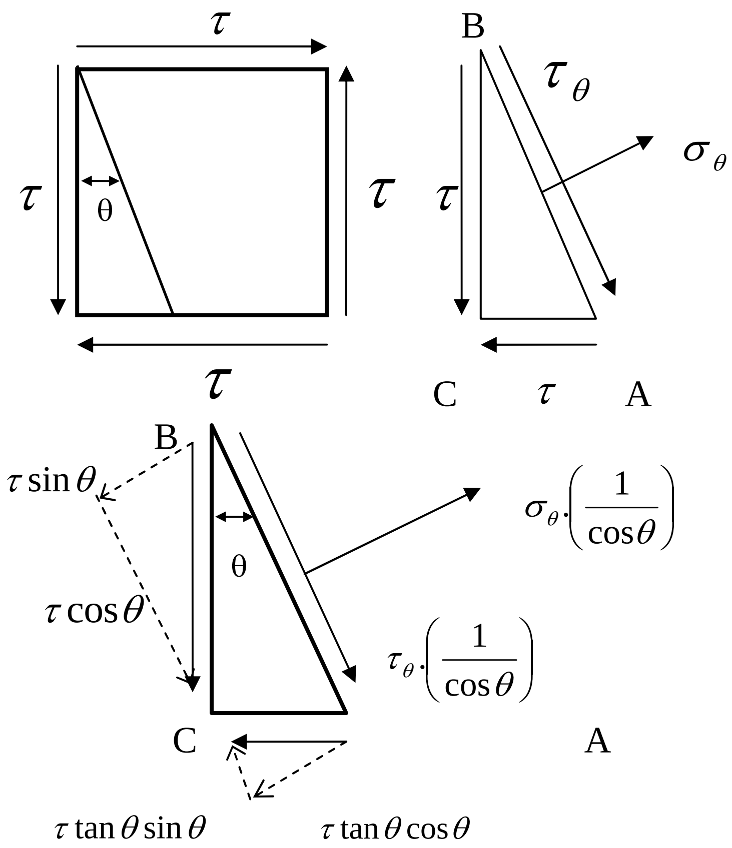
We get: -

$$\sigma_\theta = \left(\frac{\sigma_x + \sigma_y}{2}\right) + \left(\frac{\sigma_x - \sigma_y}{2}\right).\cos 2\theta$$

$$\tau_\theta = \left(\frac{\sigma_x - \sigma_y}{2}\right).\sin 2\theta$$

Case 2 Consider an element subjected to shear stresses only

Consider the same element as we did previously for the bi-axial case, but now consider 'Shear' stresses on the faces instead of 'Normal' stresses. Considering the equilibrium of forces perpendicular and parallel to the plane AB, we get:



Resolving normal to AB we have;

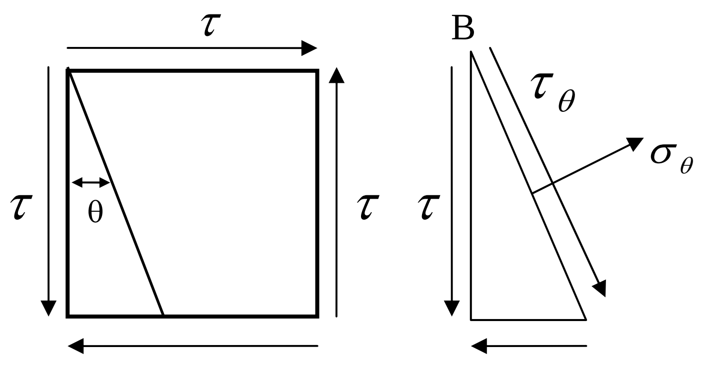
$$\sigma_\theta \cdot \left(\frac{1}{\cos\,\theta}\right) = \tau \sin\,\theta + \tau.\tan\,\theta.\cos\,\theta$$

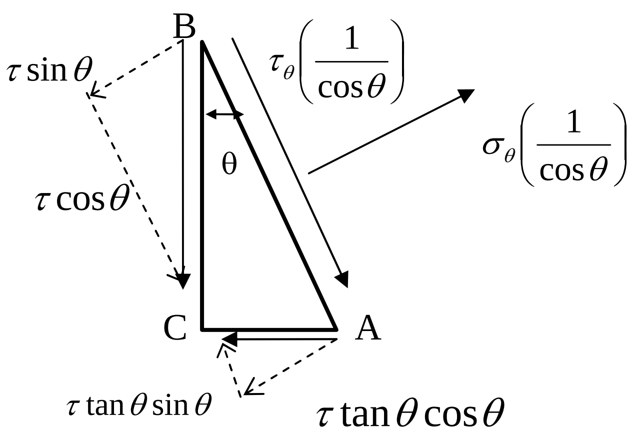
Multiplying through by cos θ we have;

$$\sigma_\theta = \tau.\sin\theta\cos\theta + \tau.\sin\theta\cos\theta$$
$$\sigma_\theta = 2\tau.\sin\,\theta.\cos\,\theta$$

$$\boxed{\sigma_\theta = \tau.\sin 2\theta}$$

Considering the force balance again – but this time resolving parallel to plane AB:





We get:

$$\tau_\theta\left(\frac{1}{\cos\theta}\right) = \tau\tan\theta.\sin\theta - \tau\cos\theta$$

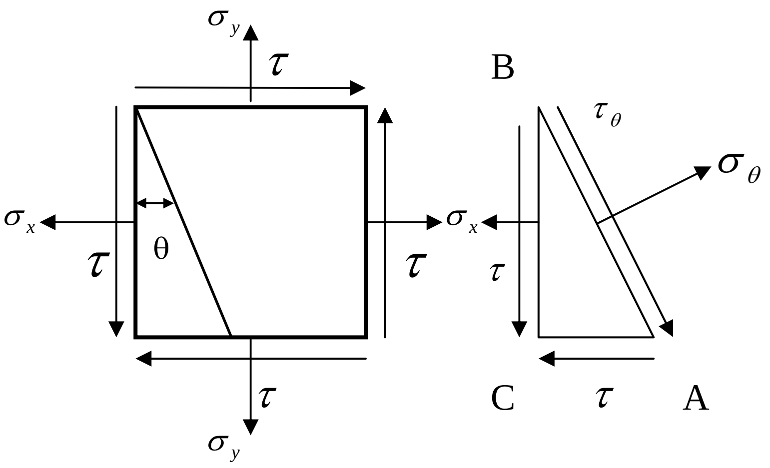
$$\tau_\theta = \tau\sin^2\theta - \tau\cos^2\theta$$

$$\tau_\theta = \tau\left(\sin^2\theta - \cos^2\theta\right)$$

$$\boxed{\therefore\ \tau_\theta = -\tau.\cos 2\theta}$$

In Summary

For the general 2D stress system shown below, the equations for normal and shear stresses on the inclined plane AB are: -



$$\sigma_\theta = \left(\frac{\sigma_x + \sigma_y}{2}\right) + \left(\frac{\sigma_x - \sigma_y}{2}\right).\cos 2\theta + \tau.\sin 2\theta$$

$$\tau_\theta = \left(\frac{\sigma_x - \sigma_y}{2}\right).\sin 2\theta - \tau.\cos 2\theta$$

Maximum and Minimum Values of Direct Stress

It is clear that as the angle of the plane AB varies, the direct stress, σ_θ will also vary. The Maximum and Minimum values of σ_θ are known as the Principal Stresses, σ_1 and σ_2.

We have derived an equation for σ_θ;

$$\sigma_\theta = \left(\frac{\sigma_x + \sigma_y}{2}\right) + \left(\frac{\sigma_x - \sigma_y}{2}\right).\cos 2\theta + \tau.\sin 2\theta$$

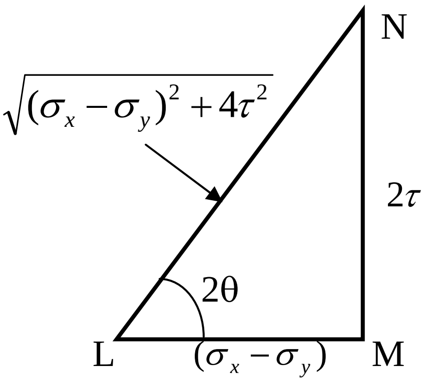
For σ_θ to be maximum or a minimum, $\dfrac{d\sigma_\theta}{d\theta} = 0$

$$0 = -\frac{(\sigma_x - \sigma_y)}{2}.2\sin 2\theta + 2\tau.\cos 2\theta$$

$$(\sigma_x - \sigma_y).\sin 2\theta = 2\tau.\cos 2\theta$$

$$\therefore \frac{\sin 2\theta}{\cos 2\theta} = \frac{2\tau}{\sigma_x - \sigma_y} = \tan 2\theta$$

Consider the triangle LMN



$$\sin 2\theta = \frac{2\tau}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}}$$

$$\cos 2\theta = \frac{(\sigma_x - \sigma_y)}{\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}}$$

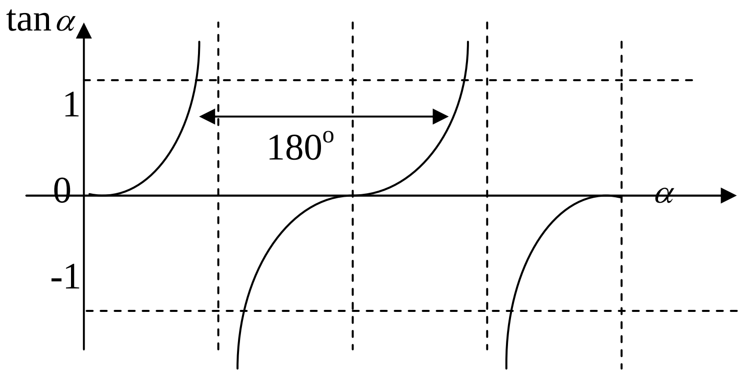
Substitution of $\sin 2\theta$ and $\cos 2\theta$ into our general equation for σ_θ, gives;

$$\boxed{\begin{aligned} \sigma_1 &= \frac{\sigma_x + \sigma_y}{2} + \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2} \\ \sigma_2 &= \frac{\sigma_x + \sigma_y}{2} - \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2} \end{aligned}}$$

Examination of the tangent curve shows that the solutions to the equation:

$$\tan 2\theta = \frac{2\tau}{\sigma_x - \sigma_y}$$

are spaced at 180^o intervals. Thus, the two planes on which the direct stress is either a maximum or a minimum are at right angle to each other.



Hence from:

$$\tan 2\theta = \frac{2\tau}{\sigma_x - \sigma_y}$$

One solution for θ is:

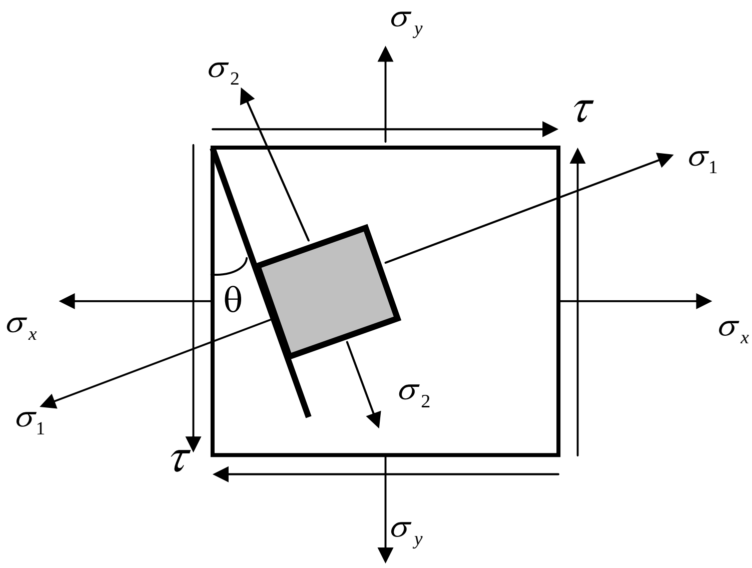
$$\theta = \frac{1}{2}\left[\tan^{-1}\left(\frac{2\tau}{\sigma_x - \sigma_y}\right)\right]$$

And the next is:

$$\theta = \frac{1}{2}\left[\tan^{-1}\left(\frac{2\tau}{\sigma_x - \sigma_y}\right) + 180^0\right]$$

$$\therefore \theta = \left[\frac{1}{2}\tan^{-1}\left(\frac{2\tau}{\sigma_x - \sigma_y}\right) + 90^0\right]$$

Thus, the planes of maximum and minimum stress are mutually perpendicular. One of these stresses acts on one plane and the other stress acts on the perpendicular plane.



It is interesting to note that the equation:

$$\tan 2\theta = \frac{2\tau}{\sigma_x - \sigma_y}$$

can be obtained from our general equation:

$$\tau_\theta = \frac{\sigma_x - \sigma_y}{2}.\sin 2\theta - \tau.\cos 2\theta$$

… by simply putting $\tau_\theta = 0$

Thus, on the planes where the principal stresses can be found there are NO shear stresses. These planes are called "Principal Planes" and the direct stresses acting on them are " Principal Stresses". Thus, at any point in a stressed material there are always mutually perpendicular planes upon which the stresses are wholly tensile or compressive and these are respectively the greatest and the least direct stresses at that point.

The Maximum Shear Stress

By means of a similar analysis the conditions for the shear stress to be maximum can be determined. Consider:

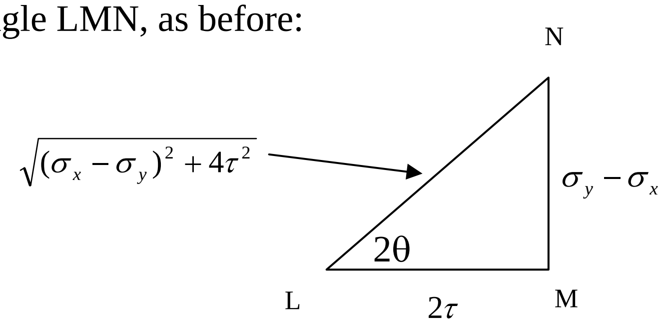
$$\tau_\theta = \frac{\sigma_x - \sigma_y}{2} . \sin 2\theta - \tau . \cos 2\theta$$

$\tau_{\max}$ is given on the plane when $\dfrac{d\tau_\theta}{d\theta} = 0$

$$\therefore (\sigma_x - \sigma_y) \cos 2\theta + 2\tau \sin 2\theta = 0$$

$$\therefore \tan 2\theta = \frac{\sigma_y - \sigma_x}{2\tau}$$

Considering the triangle LMN, as before:



Substitution of this condition gives the following for $\tau_{\max}$:

$$\tau_{\max} = \frac{1}{2}\sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2}$$

The product of equations: -

$\tan 2\theta = \dfrac{(\sigma_y - \sigma_x)}{2\tau}$ which can also be written as $\tan 2\theta = \dfrac{-(\sigma_x - \sigma_y)}{2\tau}$

and $\tan 2\theta = \dfrac{2\tau}{\sigma_x - \sigma_y}$ is –1.

This confirms that the angle separating these two conditions for 2θ is 90°,which means that the planes on which the shear stress is a maximum lie at 45° to the plane on which σ_1 and σ_2 occur.

Mohr's Stress Circle

Consider again the two general equations:

$$\sigma_\theta = \left(\frac{\sigma_x + \sigma_y}{2}\right) + \left(\frac{\sigma_x - \sigma_y}{2}\right).\cos 2\theta + \tau.\sin 2\theta$$

$$\tau_\theta = \left(\frac{\sigma_x - \sigma_y}{2}\right).\sin 2\theta - \tau.\cos 2\theta$$

If $\frac{\sigma_x + \sigma_y}{2} = A$ and $\frac{\sigma_x - \sigma_y}{2} = B$

Then these equations may be written as:

$$\sigma_\theta - A = B\cos 2\theta + \tau \sin 2\theta$$
$$\tau_\theta = B\sin 2\theta - \tau\cos 2\theta$$

Squaring and adding these gives:

$$(\sigma_\theta - A)^2 + \tau_\theta{}^2 = B^2 + \tau^2$$

This is a equation of a circle with centre at x=A, y=0 and radius:

$$r = \sqrt{B^2 + \tau^2}$$

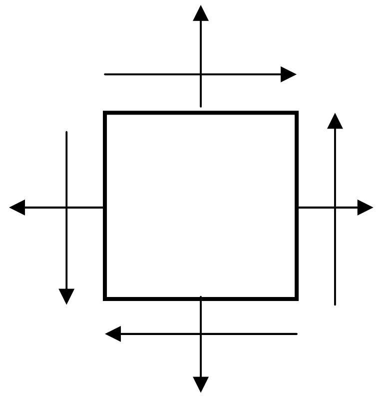
Thus the stresses τ_θ *and* σ_θ , on a plane may be graphically represented as a circle. This is the basis of the " MOHR'S STRESS CIRCLE", which is widely used in most industries.

Construction of Mohr's stress circle

It is necessary to adopt a sign convention, for the stresses:

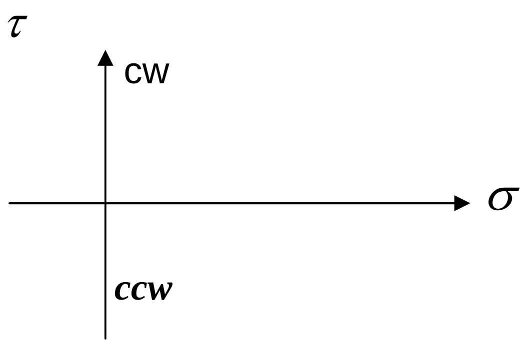
a) Direct stress :- + ve when tensile
- ve when compressive

b) Shear stress:- + ve when tending to rotate the element CW
- ve when tending to rotate the element ACW

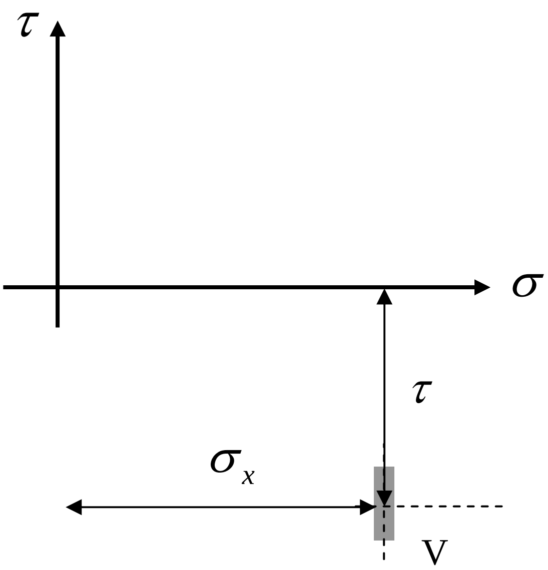


The stages of the construction are illustrated below:

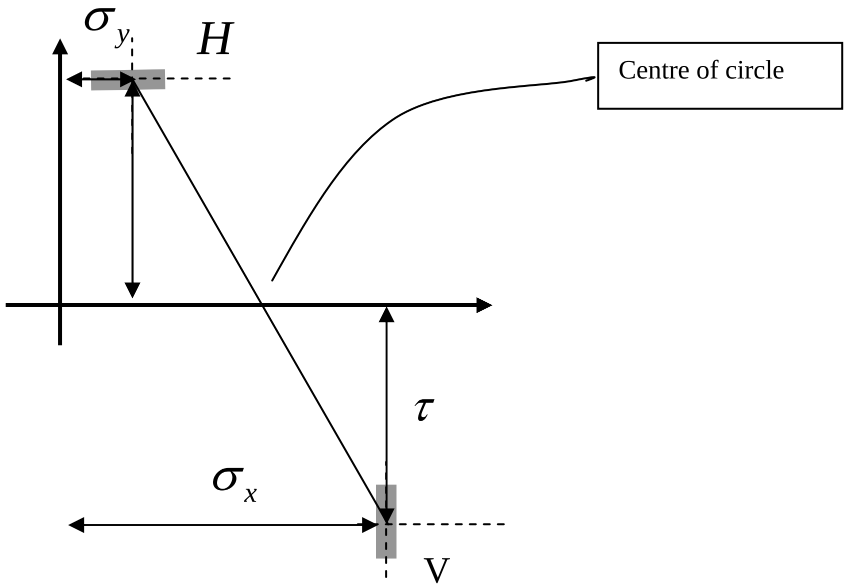
STEP 1: Draw a set of axes, which at least cover the range of stresses acting on the stressed element. The Horizontal axis represents " DIRECT STRESS", and the vertical axis " SHEAR STRESS". Note, both axes must have the same scale.



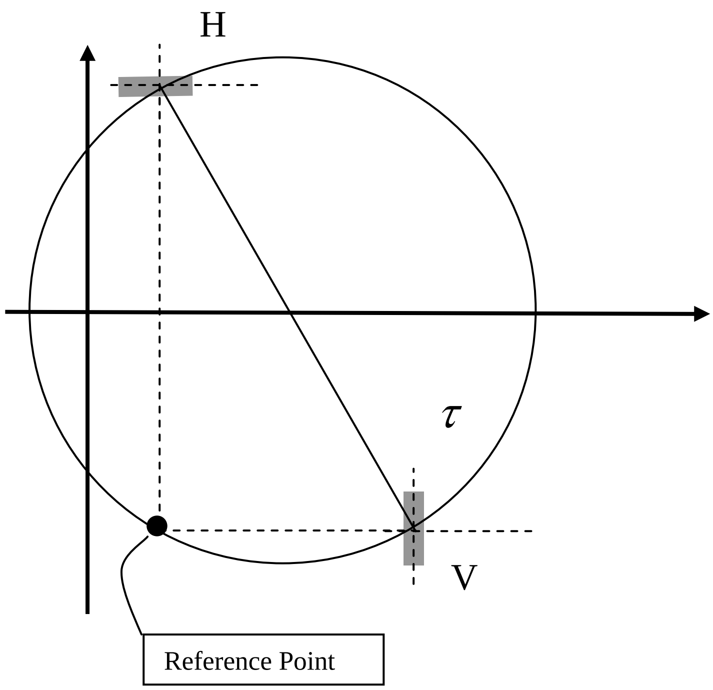
STEP 2: Plot on these axes a point which represents the stresses on the vertical face, label this point V.



STEP 3: Plot another point, which represents the stresses on the horizontal face, label this point H, and connect these points. It is helpful to draw a piece of vertical face at V and a piece of horizontal face at H. Note: for this example $\sigma_x > \sigma_y$



STEP 4: Construct a circle with VH as a diameter. This is simply done by using the intersection of VH with σ axis, as the centre. Draw lines (shown dashed) PERPENDICULAR to the small bits of surface drawn (shown shaded). Where these construction lines intersect with the circle is known as the Reference Point (or Pole Point).

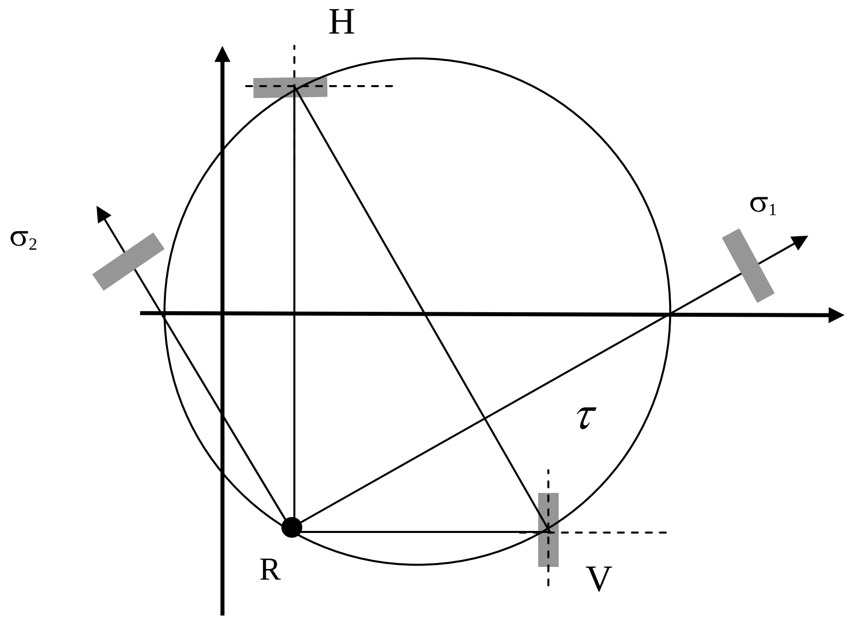


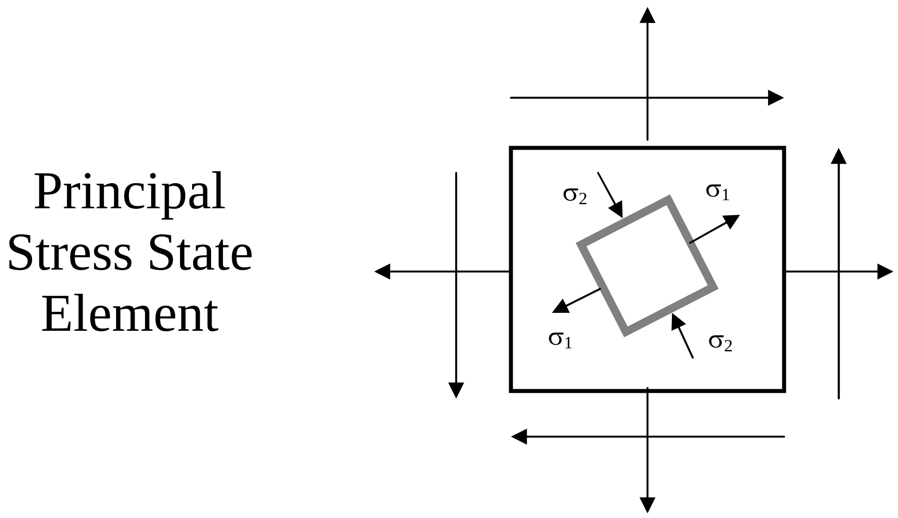
Interpretation of Mohr's Circle for Stress

The principal stresses, σ_1 (the largest) and σ_2 (the smallest) may now be determined.

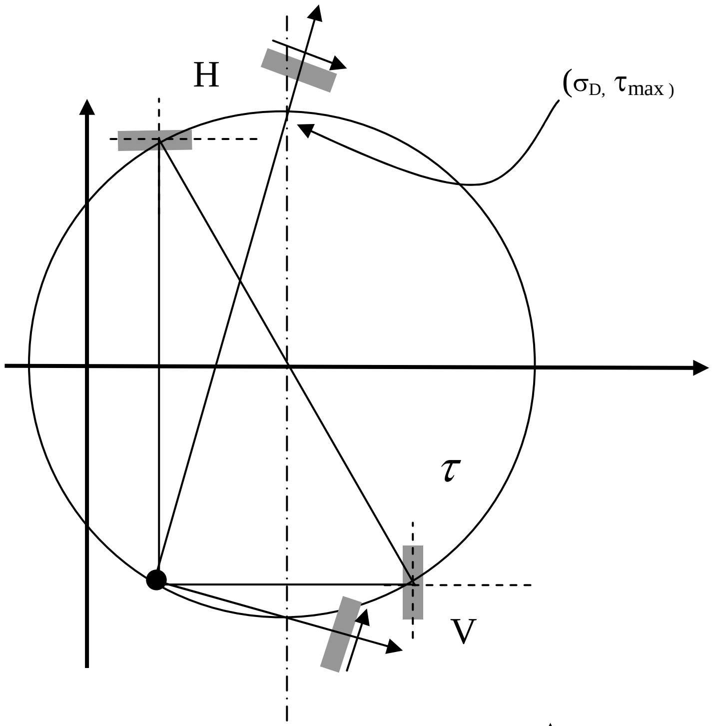
Remember: σ_1 and σ_2, are the maximum and minimum direct stresses in the element and occur on planes where the shear stress is zero.

Thus, they occur at the intersection of the circle with the direct stress axis. The direction of the principal stresses are given by lines joining R to the stress points on the circle. The planes on which the direct stresses act are called the Principal Planes.

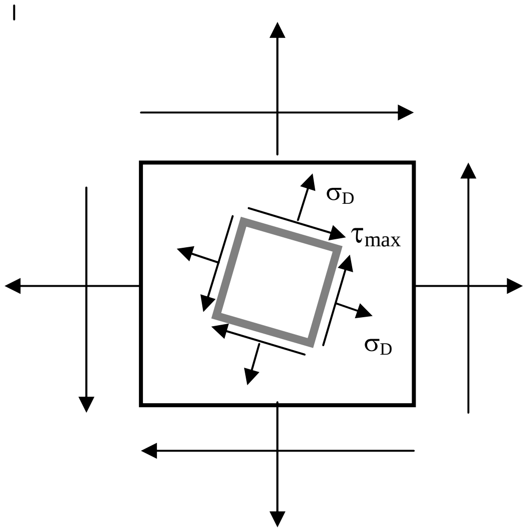




The maximum shear stresses are found in a similar fashion, except that in this situation direct stresses occur as well.

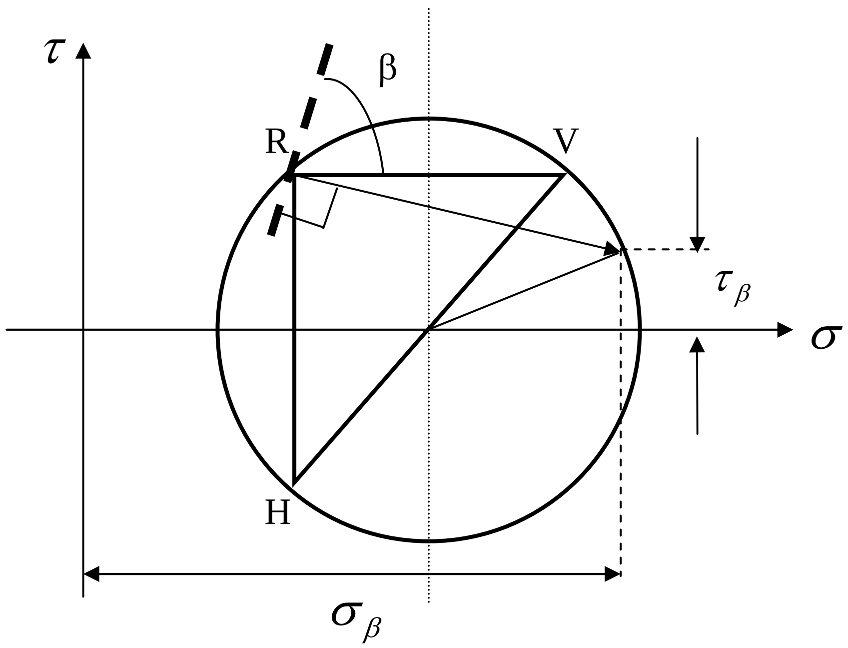


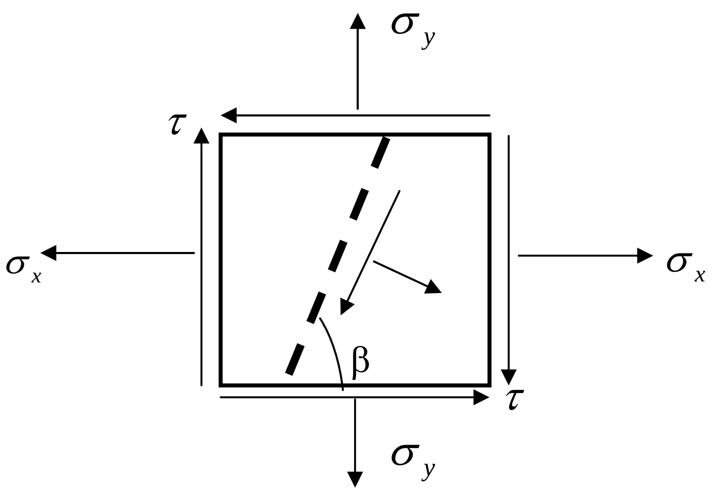
Maximum Shear
Stress State
Element



Determination of stresses on a defined plane

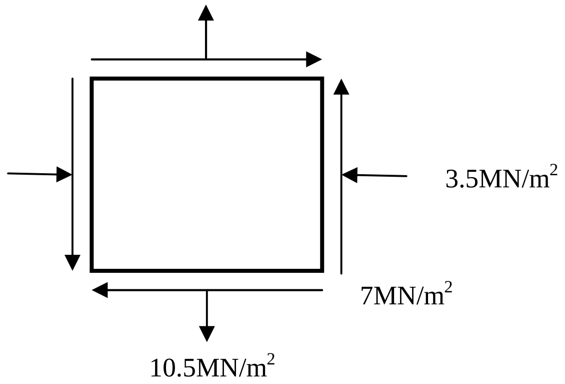
It is often desirable to determine the stresses on a given plane at some defined angle in the original stress element. This is done by placing a plane at the required angle (β ACW from +ve x axis as shown) through the reference point, R, then by projecting a normal through R until it intersects the circle. This point defines the state of stress on the plane. These values can be verified by calculation as long as the correct angle θ is used in the equations such that: $\theta = 90 + \beta$.

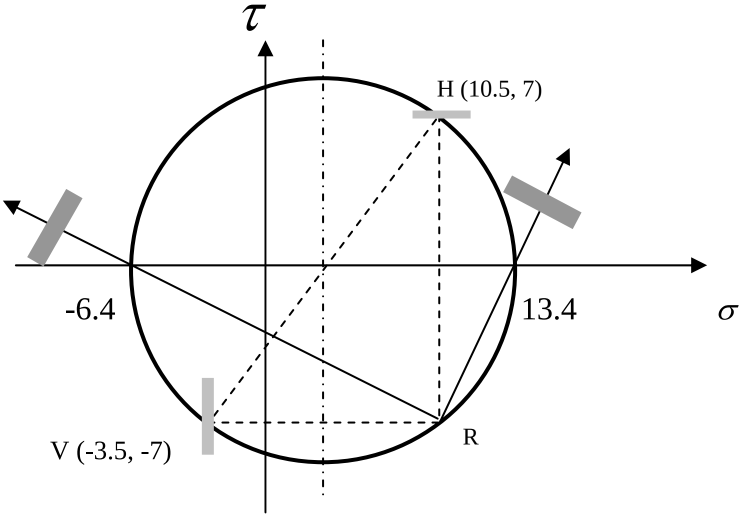




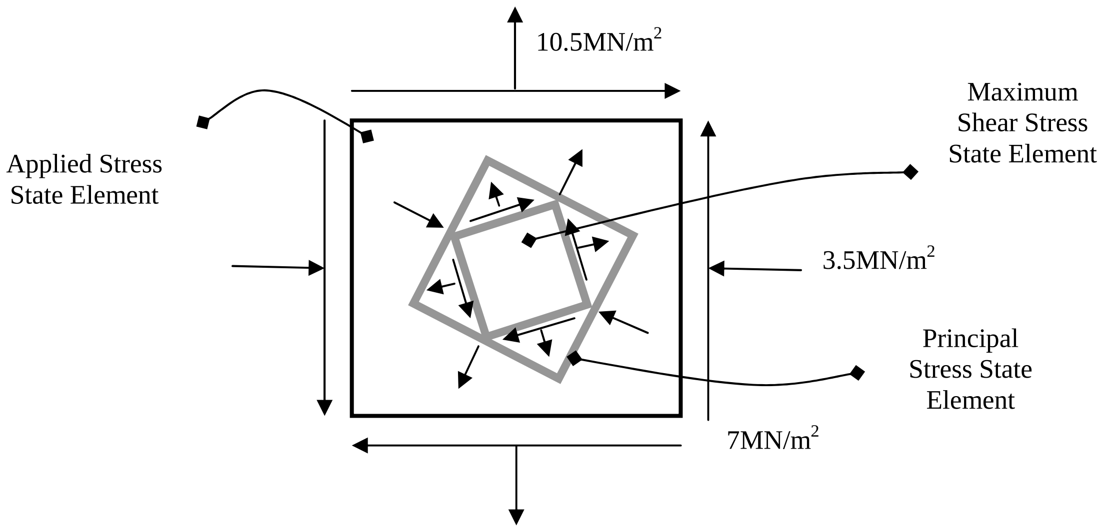
Example 1

Determine, using Mohr's circle construction, the principal stresses, maximum shear stresses and the direct stresses on these planes. Show the positions of these planes on diagrams of the element.





$$\sigma_1 = 13.4, \sigma_2 = -6.4, \tau = 9.9, \sigma_{direct} = 3.5 \text{ (MN/m}^2\text{)}$$



Example 2

A solid shaft subjected to an applied torque and axial compressive force is to be designed using the following criteria.

Criteria (to be taken together):

i. The shaft must be able to transmit 2 megawatts at 1000rpm.

ii. The applied axial compressive stress due to the axial force must not exceed 250N/mm^2.

iii. The maximum permissible shear stress due to the combined axial loading and torque is to be 200N/mm^2.

Determine the minimum necessary shaft diameter and the magnitude and direction of the maximum compressive stress present in the material.

Design criteria i

Transmits 2×10^6 watts of power at 1000rpm – from this we need the torque.

From

$$\text{Power} = \text{T} \times \text{w}$$

$$2 \times 10^6 = \text{T} \times \frac{2\pi}{60} \times 1000$$

$$\underline{\therefore T = 19098.59 Nm}$$

$$1000 \text{ rpm} = 1000 \left[\frac{revs}{\min}\right]\left[\frac{\min}{60\sec}\right]\left[\frac{2\pi rads}{revs}\right]$$

From this applied torque we can get the Shear Stress, τ

Using:

$$\frac{T}{J} = \frac{\tau}{r} \qquad \therefore \tau = \frac{T.r}{J}$$

Where: $J = \dfrac{\pi D^4}{32}$ r=D/2 T as above

$$\therefore \tau = \frac{19098.59 \times (D/2)}{\left(\dfrac{\pi D^4}{32}\right)}$$

$$\therefore \tau = \frac{97268.33}{D^3} N/m^2$$

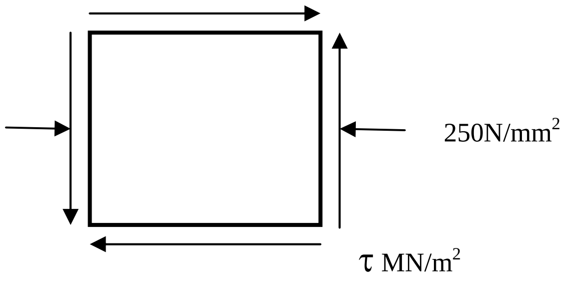
Design criteria ii

Applied axial compressive stress, due to the axial force, must not exceed 250N/mm^2.

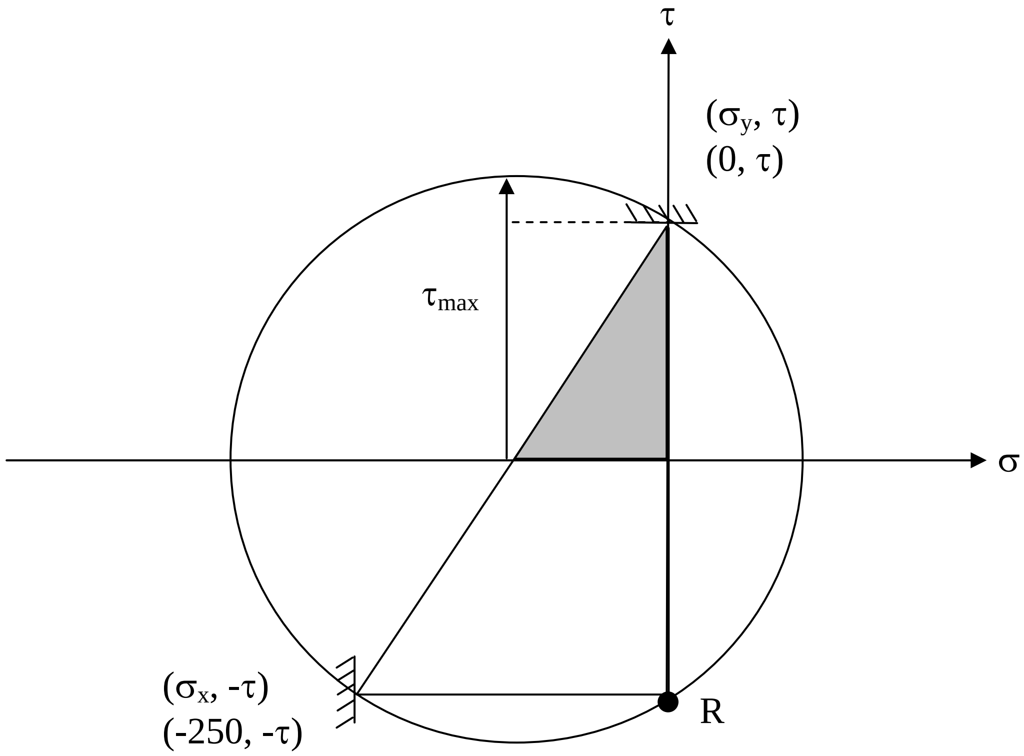
Design criteria iii

τ_{max} = 200N/mm^2

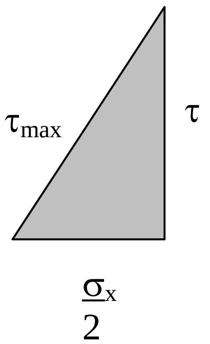
So, for an element on the surface of the shaft:



Solve by SKETCHING Mohr's Circle.



Consider the shaded triangle.



From Pythag:

$$\tau_{max}^2 = \tau^2 + (\sigma_x/2)^2$$

$$\therefore \tau = \sqrt{200^2 - (-125)^2}$$

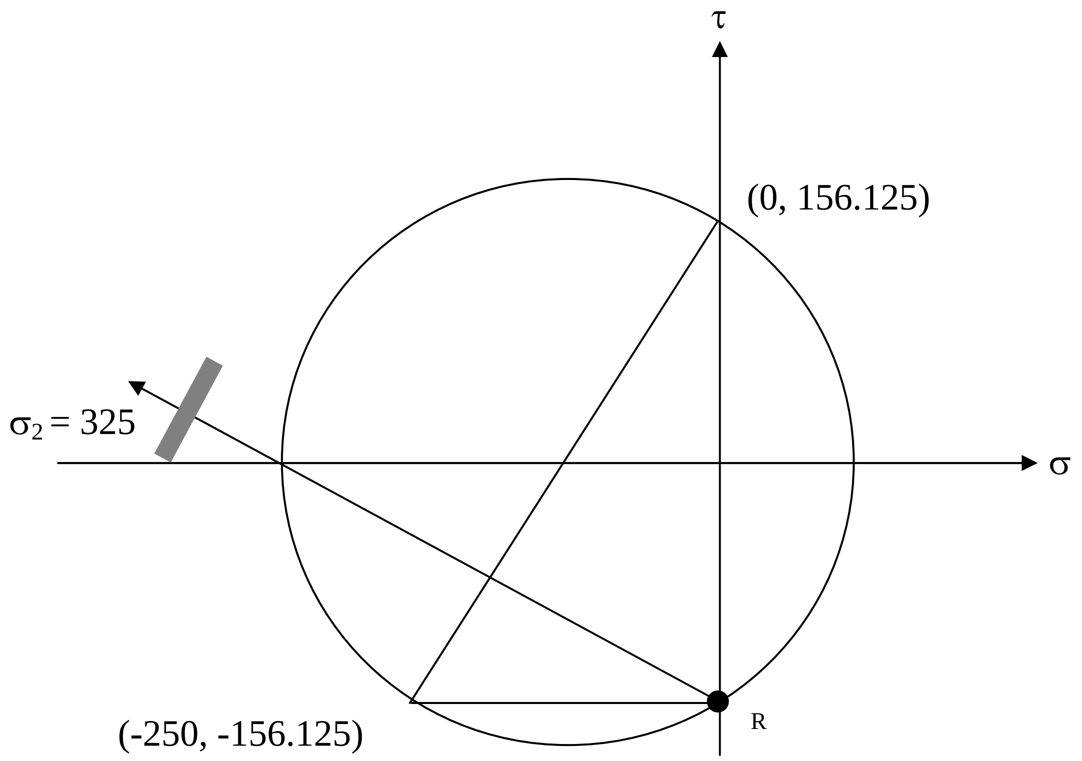
$$\therefore \tau = 156.125 MN/m^2$$

Hence, knowing:

$$\therefore \tau = \frac{97268.33}{D^3} N/m^2$$

Then, we can find D: $D = \sqrt[3]{\dfrac{97268.33}{156.125 \times 10^6}} = 85.4mm$

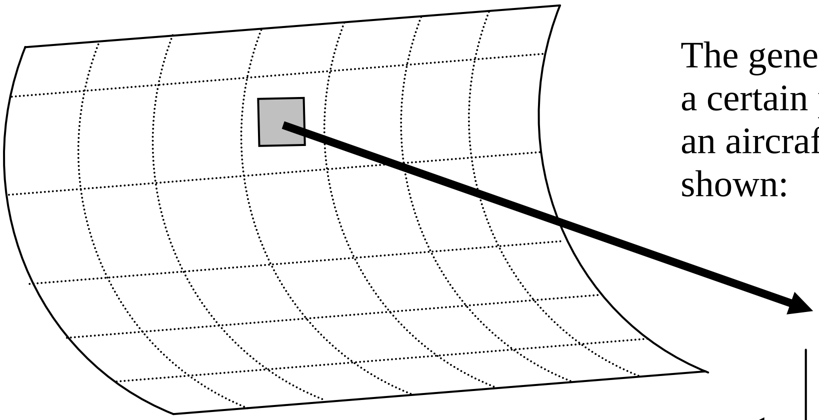
Finally, we can find the magnitude and direction of the maximum compressive stress present in the material:



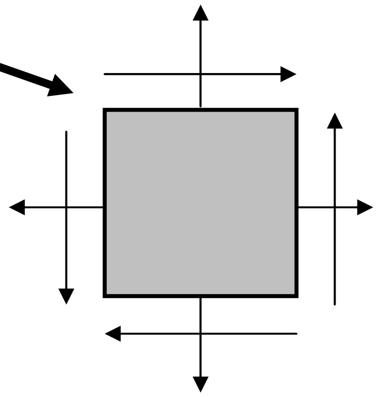
So, $\sigma_2 = 325$ MN/m^2 Compressive

Tutorials

Question 1



The general state of stress at a certain point on the skin of an aircraft fuselage is as shown:



Assume values for the applied stress system and then determine the values of the principal stresses and maximum shear stress.

Try varying these assumed values and explore what happens to the position of the Mohr's cricle and the magnitude of key values as a result.

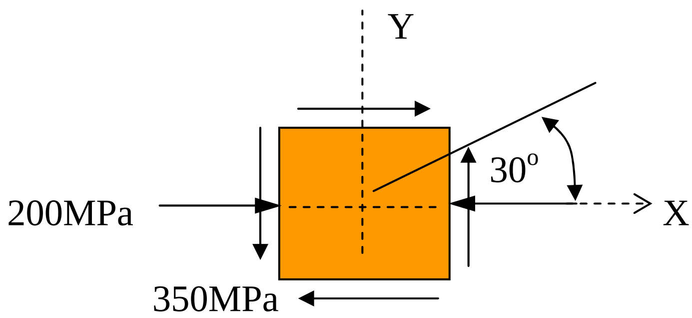
Question 2

The state of plane stress at a point on the upper surface of an aircraft wing is shown on the stress state element shown:

i) Using Mohr's Circle for stress, calculate the principal stresses in the system.

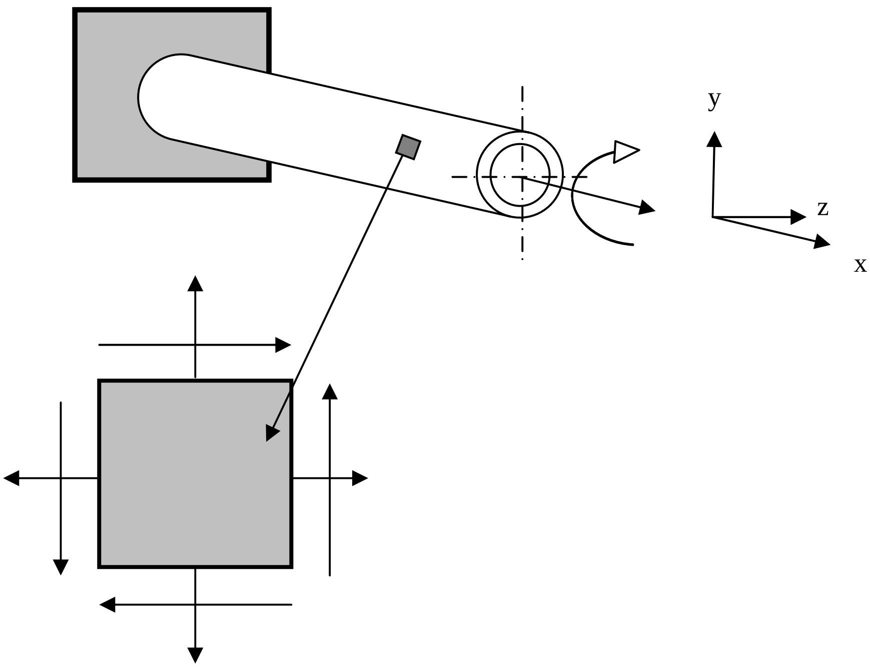
ii) Calculate the maximum in plane shear stress.

iii) The wing skin is fabricated from two 5mm thick alloy plates riveted together along a path which forms an angle of 30^o counter clockwise with the X axis. Using either Mohr's circle for stress or an analytical method, determine the stresses exerted on the element obtained by rotating the element counter clockwise through 30^o to line up with the rivet seem/line.



Question 3

A bar of outside diameter 60mm and inside diameter 40mm is subjected to a torque about its longitudinal axis of 120Nm and to a tensile load of 80kN, as shown.



Find:

a) The applied tensile stress present in the bar cross-section due to the tensile load;

b) The torsional shear stress present in the outer fibres of the bar due to the applied torque;

c) Using a relevant plot of a Mohr's circle (drawn to scale), find:

   i) the magnitude and direction of the principal stresses present in the outer fibres of the bar. Show your results on a suitable stress block diagram.

   ii) The angle between the direction of the maximum principal stress and the x axis.

   iii) The stresses on a plane in an element oriented at 10° anticlockwise from the x-axis. Show your results on a suitable stress block diagram.

Strength of Materials

When dealing with the design of structures or components the physical properties of the constituent materials are generally found from the results of experiments which have only subjected the materials to the simplest stress conditions. The most usual test is a simple tensile test in which the value of stress at some defining point is easily determined.

The strengths of materials under complex stress systems are **not** generally known. In practice it is these more complex systems of stress which are often encountered, and it is therefore necessary to have some basis for determining allowable working stresses so that failure will not occur.

Simple Axial Loading

If a ductile metal is subjected to simple axial loading, it is found that beyond a certain point, stress is no longer proportional to strain, which results in there being a permanent deformation when the stress is removed. The material is then said to have yielded or exceeded its elastic limit. Knowing the stress at which this behaviour commenced, it would then be a simple matter to design a component from the same material to withstand a particular axial load without any yielding occurring.

This is a simple matter because IN THIS CASE there is only one principal stress to consider!

The problem of designing a more complex component such as a pressure vessel or a rotating disc (turbine blade) so that the material remains elastic is more difficult.

A number of theoretical criteria have been proposed which aim to obtain an acceptable correlation between estimated component life and that actually achieved under service load conditions for both brittle and ductile material applications.

When a tensile test specimen fails by yielding the failure occurs at a specific principal (axial) stress and strain, a maximum shearing stress of one half the axial stress and a specific amount of strain energy per unit volume stressed material.

Since all these limits are reached simultaneously for an axial load, it makes no difference which failure criterion (stress, strain or energy) is used for predicting failure in another axially loaded member of the same material.

Bi-axial and Tri-axial Loading

For an element subjected to bi-axial or tri-axial loading, however, the situation is more complicated because the limits of normal stress, normal strain, shearing stress, and strain energy are not all reached simultaneously.

In such cases it becomes important to determine the best criterion for predicting failure because test results are difficult to obtain and the combinations of loads are endless.

For all criterion the value of the critical property (named in title of criterion) is determined for both the simple tension case and the full potentially 3D complex stress system. These values are then equated to produce the criterion for failure.

Criterion for Ductile Materials

Materials which exhibit yielding followed by some plastic deformation prior to fracture as, measured under-simple tensile or compressive stress, are termed **ductile**.

This property provides a design reserve for materials if they exceed the elastic range during service. Plastic deformation is related to slip in the grain structure which is dependent on a shearing action.

Many criteria have been developed for ductile materials, but there are two in particular which are now widely accepted in most industries:

- ***The Maximum Shear Stress Theory (Tresea Criterion)***
- ***The Shear or Distortion Strain Energy Theory (von-Mises Criterion)***

Criterion for Brittle Materials

Brittleness in a material may be defined as an inability to deform plastically. Materials such as glass, some irons, concrete and some plastics, when subjected to tensile stress, will generally fracture at or just beyond the elastic limit.

We are therefore more concerned with a fracture criterion than a yield criterion. The most commonly used criterion is:

- ***The Maximum Principal Stress Theory (Rankine Criterion)***

Limitations

If it is anticipated that failure may occur in service due to fatigue, creep, buckling etc. then you should not use these criterion.

If it is assumed to be relevant to use an elastic failure approach then it is necessary to consider which of the criterion is the most appropriate for the material in question and for the service loading expected.

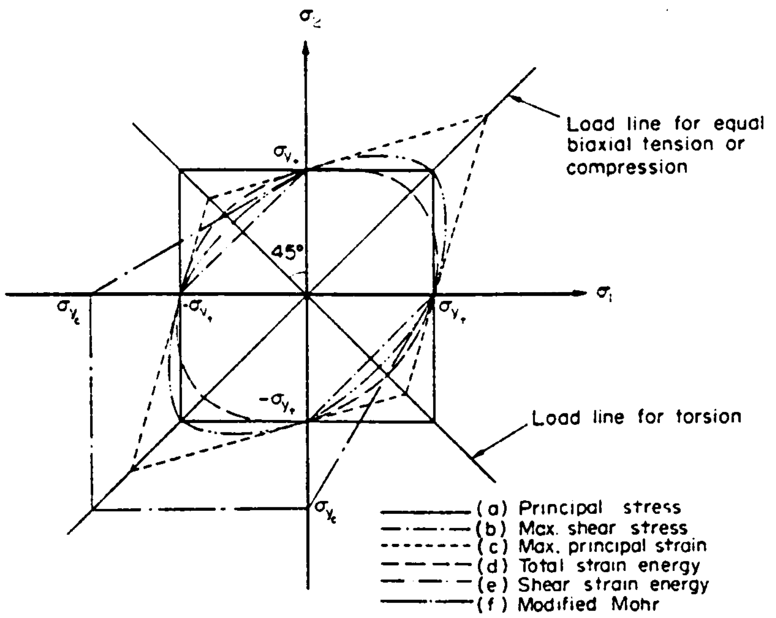
Not all ductile metals obey Tresca; some steels and alloys of aluminium are governed by a critical value of the strain energy of distortion.

For the majority of cases the Von-Mises (Distortion Strain Energy) criterion is considered to be the most reliable.

All theories produce similar results in loading situations where one principal stress is large compared to another.

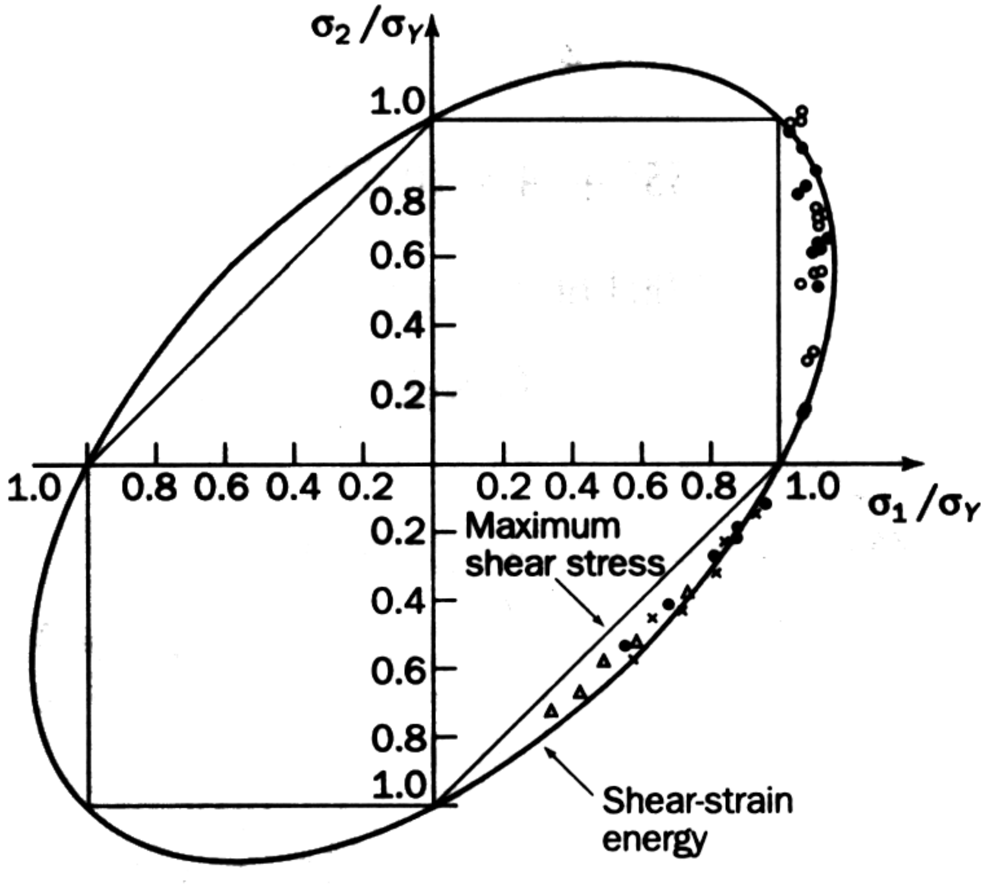
Greatest discrepancy found in second and fourth quadrants where the principal stresses are of the opposite sign but equal magnitude.

Yield Loci



Determination of Yield Loci

In practice, these graphs are based on experimental data, which does mean they are best fit lines through these data points.

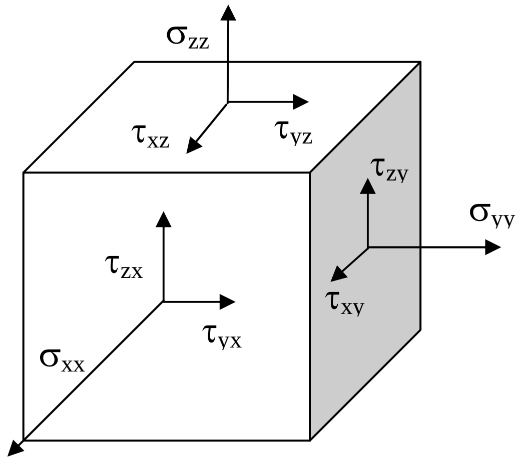


Inside any particular locus (or failure envelope) elastic conditions prevail. Points outside indicate that failure is predicted to occur.

Three Dimensional Stress

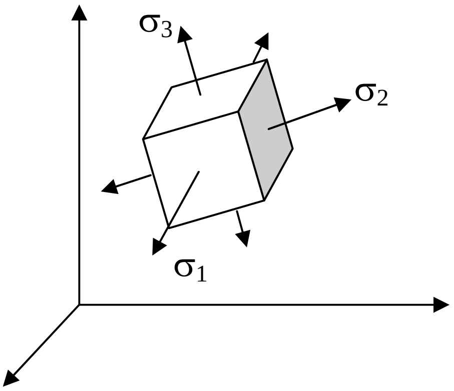
In order to make use of elastic failure criteria we need to consider how to interpret principal stress values.

Consider the general state of 3D stress at any other point in a body.

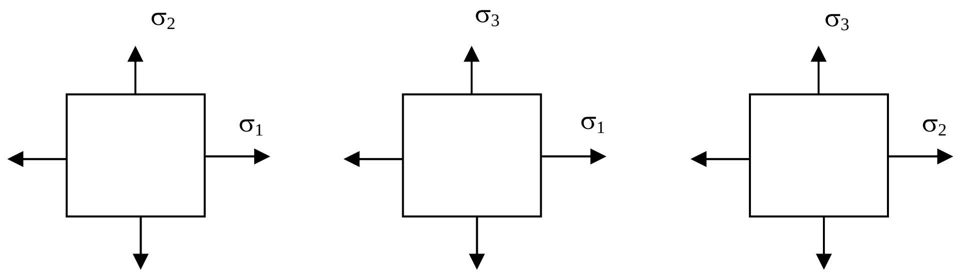


We can also show the principal element at the same point on the body. A full study of stress in three dimensions shows that there are three mutually perpendicular planes of principal stresses, each free of shear stress. It may be shown that the three planes on which the three principal stresses occur, σ_1 and σ_2, σ_2 and σ_3, σ_3 and σ_1, can all be represented for a 2D case.

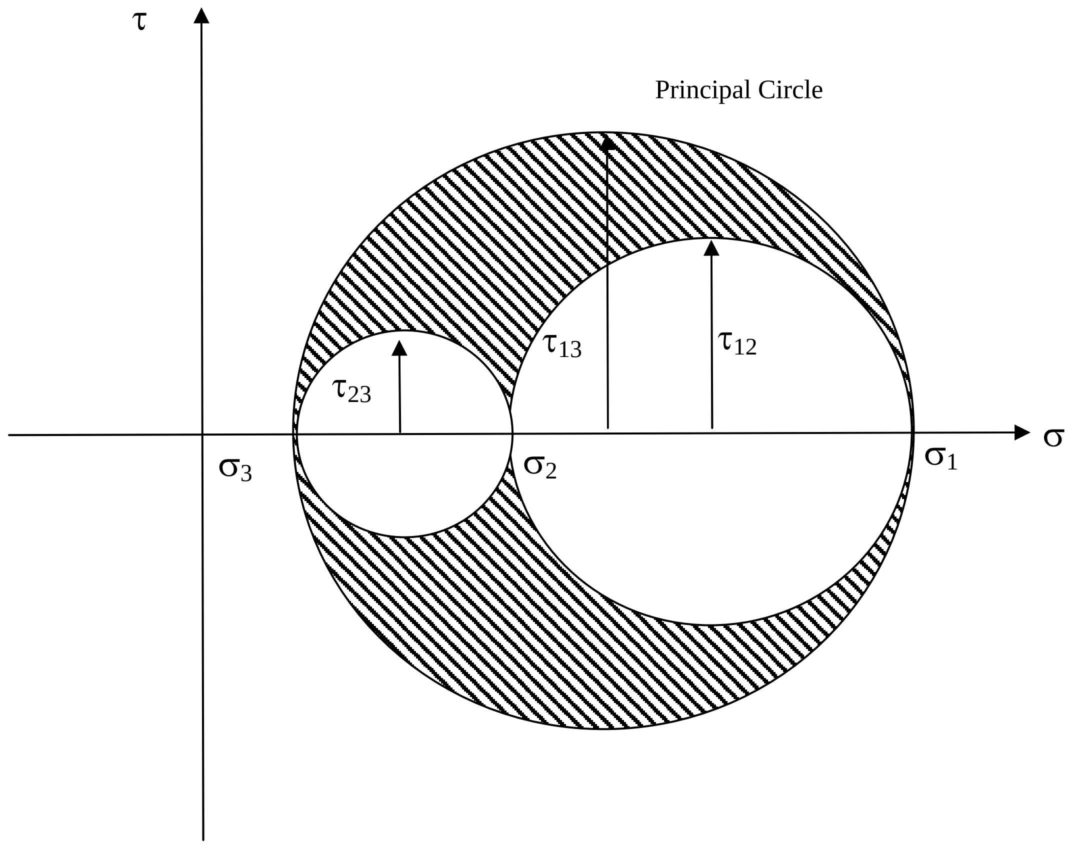
Important: - It is convention to label the principal stresses so that $\sigma_1 > \sigma_2 > \sigma_3$.



We can then look at true views on the various faces of the principal element and for each 2D stress condition a Mohr's Circle can be drawn.



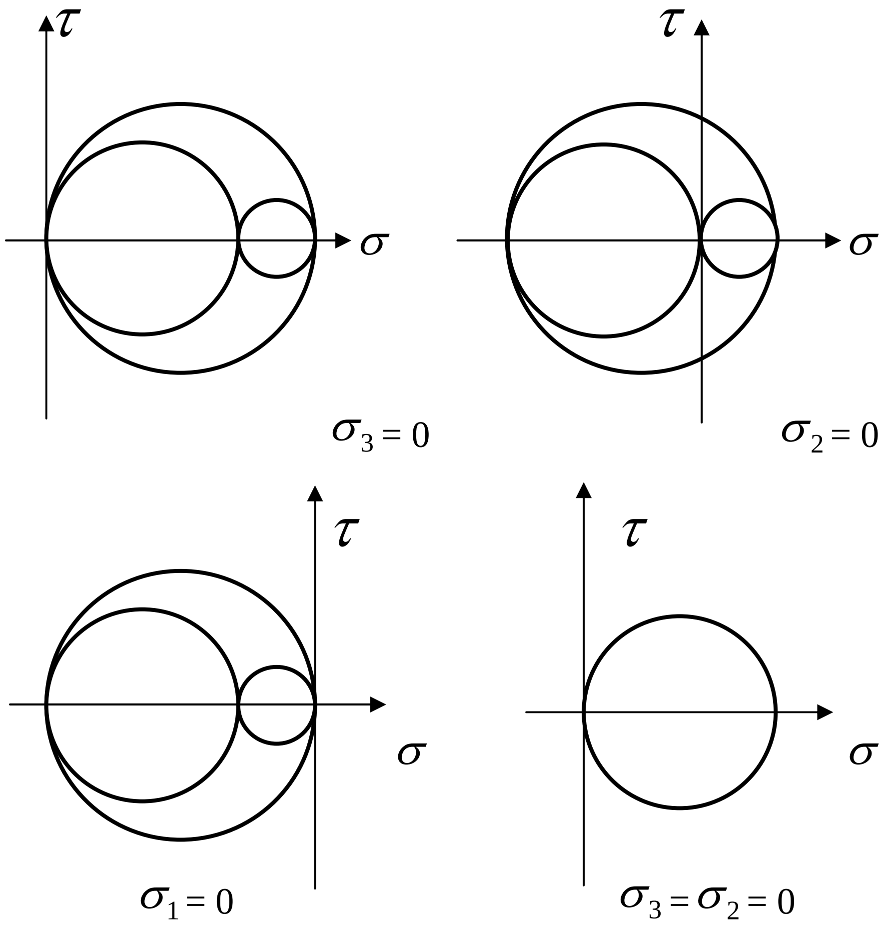
The three Mohr's circles can then be combined to produce the complete 3D Mohr's Circle. The large circle between points σ_1 and σ_3 represents stresses on all planes through the point in question containing the σ_2 axis etc.



It is important that when considering two dimensional cases that the maximum (greatest) stress must be determined by identifying which principal stress is zero and then constructing the principal circle!

Depending on the values of the non-zero principal stresses, the zero principal stress may be σ_1, σ_2 or σ_3.

Viz:



The Maximum Shear Stress Theory (TRESCA CRITERION)

This theory states that failure can be assumed to occur when the maximum shear stress in the complex stress system becomes equal to that at the yield point in the simple tensile test. Since maximum shear stress is half the greatest difference between two principal stresses the criterion of failure becomes:

$$\tfrac{1}{2}(\sigma_1 - \sigma_3) = \tfrac{1}{2}(\sigma_y)$$

$$\therefore (\sigma_1 - \sigma_3) = \sigma_y$$

i.e. the value of σ_3 being algebraically the smallest value (taking into account the sign and the fact that one stress may be zero).

For like stresses in the 2D case (i.e. both σ_1 and σ_2 tensile– the maximum shear stress criterion is:

$$\tfrac{1}{2}(\sigma_1 - 0) = \tfrac{1}{2}(\sigma_y)$$

$$\therefore \sigma_1 = \sigma_y$$

For unlike stresses the criterion is:

$$\tfrac{1}{2}(\sigma_1 - \sigma_3) = \tfrac{1}{2}(\sigma_y)$$

This criterion is often used as it is SIMPLE! Often used for ductile metals in general design and analysis work.

In General Tresca states that:

$$\tau_{\max} = \tfrac{1}{2}(\sigma_1 - \sigma_3) = \tfrac{1}{2}(\sigma_y)$$

The Shear or Distortion Strain Energy Criterion (Von Mises Criterion)

This criterion distinguishes between two forms of strain – that which changes volume and that which changes shape. It was proposed that the former played no part in the yielding process. The criterion then states that yield takes place when the distortional (shear) strain energy in a component reaches the distortional strain energy in a tensile test at yield.

Volumetric strain energy/unit volume = $\dfrac{3(1-2\nu)\sigma_m{}^2}{2E}$

Where $\sigma_m = \tfrac{1}{3}(\sigma_1 + \sigma_2 + \sigma_3)$

Shear Strain Energy/Unit Volume = Total Strain Energy/Unit Volume – Volumetric Strain Energy/Unit Volume

$$= \left[\frac{1}{2E}\left(\sigma_1{}^2 + \sigma_2{}^2 + \sigma_3{}^2\right) - \frac{\nu}{E}\left(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_1\sigma_3\right)\right] - \frac{(1-2\nu)(\sigma_1 + \sigma_2 + \sigma_3)^2}{6E}$$

$$= \frac{(1+\nu)}{6E}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right]$$

The value of the shear strain energy/unit volume for the tensile test is obtained by substituting $\sigma_1 = \sigma_y$, $\sigma_2 = \sigma_3 = 0$, this gives:

Shear strain energy per unit volume = $\dfrac{(1+\nu)}{3E}\sigma_y{}^2$

Thus the condition for yielding is:

$$\frac{(1+\nu)}{6E}\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right] \geq \frac{(1+\nu)}{3E}\sigma_y{}^2$$

$$\boxed{\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right] \geq 2\sigma_y{}^2}$$

Failure of Fibrous Composite Materials

When dealing with isotropic materials it has been possible to develop a failure criterion on the basis of a limiting parameter (yield or proof stress). For fibre reinforced composites it is not possible to define failure in terms of a single parameter.

The strength of a laminate can be determined by the strength of its weakest component part. The strength of a laminate will therefore be determined by the individual plies within the laminate.

One of the most popular criteria for laminates is the Tsai-Hill criterion which is based on the von-Mises criterion which was expanded to anisotropic materials by Hill and applied to composites by Tsai.

The criterion may be expressed as:

$$\left(\frac{\sigma_x}{T_x}\right)^2 - \left(\frac{\sigma_x \sigma_y}{T_x^{\,2}}\right) + \left(\frac{\sigma_y}{T_y}\right)^2 + \left(\frac{\tau_{xy}}{S_{xy}}\right)^2 = 1$$

Where: σ_x = stress parallel to the fibres
σ_y = stress perpendicular to the fibres
T_x = tensile strength parallel to the fibres
T_y = tensile strength perpendicular to the fibres
τ_{xy}, S_{xy} = shear stress and shear strength values

An alternative version, written in terms of a safety factor, is shown below:

$$\text{S.F.} = \frac{T_x}{\sqrt{\sigma_x^{\,2} - \sigma_x \sigma_y + \dfrac{T_x^{\,2}}{T_y^{\,2}}\sigma_y^{\,2} + \dfrac{T_x^{\,2}}{S_{xy}^{\,2}}\tau_{xy}^{\,2}}}$$

Factors of Safety

A Factor of Safety (or factor of ignorance) is used to cover things like an allowance for variations in material properties, assumed loading and support conditions etc.

Points to consider are:

Although many designs are based on static loading conditions designs are often subjected to fluctuating, dynamic or impact loading which may significantly increase stress levels.

The mechanical properties of many components is often not known exactly. Assumptions are therefore made and values used form standard texts and reference manuals.

Assumptions about a material being homogenous and isotropic (same in all directions) are rarely true due to flaws, inclusions or other weaknesses in the material.

Effect of manufacturing technique used, residual stresses from welding etc. may reduce the service strength of a component.

Engineering components can be very complex requiring expensive and time consuming analysis, but the analysis is often still based on numerous assumptions and can lead to significant errors compared to the actual situation.

It is because of the uncertainties of loading conditions, design procedures, production methods, etc. that an appropriate factor of safety is required. This factor is normally based on the yield stress of a material:

Factor of safety = Yield Stress / Allowable Working Stress

Factors of safety vary considerably from one branch of engineering to another. The selection of a factor of safety is based on a consideration of the social, human safety and economic consequences of a failure. Typical values range from 2.5 (for relatively low consequence, static load cases) to 10 (for shock load and high safety risk applications).

Many industries, such as aerospace and automotive, continually strive to trim these reserve factors in an attempt to drive down production (material) costs and weight!

Example 1

In a design test case the three principal stresses “in the correct order” are:

$\sigma_1 = 13.4\text{MN/m}^2$, $\sigma_2 = 0$, $\sigma_3 = \text{-}6.4\text{MN/m}^2$

Determine the factor of safety, which exists on the yield stress, if a material with a yield stress of 50MN/m^2 is being considered. Use the Tresca Yield Criterion.

According to Tresca, at the onset of yield: $\sigma_y / F = (\sigma_1 - \sigma_3)$

Where ‘F’ is the Factor of Safety

Hence, 13.4 –(-6.4) = 50/F

$\therefore$ <u>F = 2.525</u>

Example 2

A material subjected to a simple tension test shows an elastic limit of 240MN/m^2. Calculate the factor of safety provided if the principal stresses set up in a complex 2D stress system are limited to 140MN/m^2 tensile and 45MN/m^2 compressive. The appropriate failure criteria are:

a) the maximum shear stress theory
b) the maximum shear strain theory

a) TRESCA states that: $(\sigma_1 - \sigma_3) = \sigma_y$

Here we have a 2D system i.e. one principal stress is zero! ($\sigma_2 = 0$)

So we can write: $\sigma_1 = 140\text{MN/m}^2$, $\sigma_2 = 0$, and $\sigma_3 = -45\text{MN/m}^2$

With a factor of safety (FOS) applied to σ_y , we have:

$$\frac{\sigma_y}{FOS} = \sigma_1 - \sigma_3 \qquad \frac{240}{FOS} = 140 - (-45)$$

$$\therefore FOS = \frac{240}{185} = 1.3$$

<u>So, the FOS required is 1.3</u>

b) Von-Mises states that: $\left[(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2\right] = 2\sigma_y{}^2$

With the principal stress values used above:

$$\left(\frac{240}{FOS}\right)^2 = \tfrac{1}{2}\left((140-0)^2 + (0-(-45))^2 + (-45-140)^2\right)$$

$$\therefore FOS^2 = 2.063$$

<u>So, the FOS required is 1.44</u>

Airworthiness Regulations – Loads and Factors

The control of weight is of extreme importance. Designing an aircraft for loads greater than the aircraft experiences will obviously add considerable weight and decrease its performance.

To ensure minimum standards of strength and safety airworthiness regulations define several factors which the structure of the aircraft must satisfy.

Limit load - The maximum load anticipated under normal operating conditions.

Proof load - The product of the *limit load* and the *proof factor* (1-1.25) of safety.

Ultimate load - The ultimate load is the product of the *limit load* and the *ultimate load factor* (usually 1.5) of safety.

In general the Ultimate Factor is 1.5 for both military and civil aircraft, but must be-checked for individual components, notably the factor for some aspects of landing gears and hydraulic systems is 2.0.

Proof and Ultimate Factors of Safety are used to allow for loads greater than those anticipated in normal operating conditions, (maximum gust condition) and to a lesser extent for poor assumptions, variation in material strengths, a guard against early fatigue initiation.

The Proof Factor is usually 3/4 of the Ultimate Factor for military aircraft (i.e. 1.125) and 2/3 of the Ultimate Factor for civil aircraft.

The structure must be capable of carrying the Proof Load without detrimental distortion and should not fail until the Ultimate Load has been achieved.

An aircraft is subjected to a variety of loads during its operational life – these include:

- manoeuvre loads
- undercarriage loads
- cabin pressure loads
- gust loads
- buffeting and induced vibrations

The first three are generally controlled and known. The latter two depend on atmospheric conditions and are based on in-flight data. Statistical predictions made based on data available.

The aim would be for zero (or negligible) occurrence of the ultimate load and an extremely low rate of occurrence of the proof load.

The limit load may be produced by a specified manoeuvre or by an encounter with a particularly severe gust.

Stress

Proof Stress is the stress which when applied to a tensile test specimen and then - removed will not have caused any permanent extension in the specimen greater than the specified proof stress percentage e.g. 0.1% or 0.2% proof stress (extension).

Ultimate Stress is the maximum nominal tensile stress in a specimen at fracture. The load to fail the specimen being related to the original cross-sectional area of the specimen.

The above definitions relate to metallic materials, as will be seen when considering composites, the categorisation of the strength in this manner is insufficient

(Allowable) Permissible Design Stress - This is the maximum stress that is allowed when the ultimate load is applied. It is not necessarily the ultimate stress of the material, as other criterion, notably buckling, or a feature such as a lug or joint may limit the stress that can be applied.

Reserve Factor (or margin of safety in America, being RF -1) is used to express the amount of reserve strength in a particular design feature.

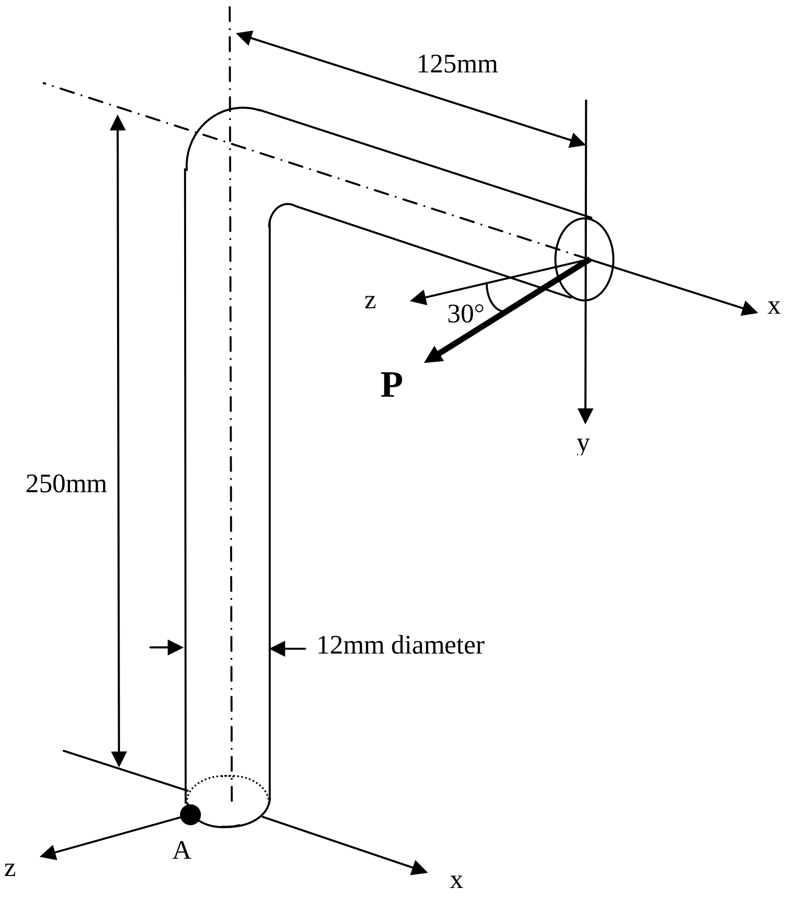
Tutorials

Question 1

Part of a supporting bracket consists of a steel rod of 12mm diameter, fixed at its lower end and containing a right angle bend which lies in the xy plane, as shown.

The force P applied at the free end lies in the yz plane and is inclined at 30° to the z axis, as shown. The 0.1% proof stress for yielding in simple tension of the steel is $200 MN/m^2$.

Calculate the value of P which will cause the onset of yielding at point A on the outer surface according to the Shear Strain Energy Criterion.



Question 2

A steel tube forms part of an aircraft undercarriage. The tube has a thickness of 3mm and a mean diameter of 100mm. Calculate the torque, which can be transmitted by the tube with a factor of safety of 2.25, if the criterion of failure is:

a) maximum shear stress (Tresca)
b) maximum shear strain energy (von-Mises)

The elastic limit of the steel in tension is 225MN/m^2 and Poisson's Ratio is 0.3.

Question 3

A circular steel cylinder of wall thickness 10mm and internal diameter 200mm is subjected to a constant internal pressure of 15MN/m^2. Determine how much:

a) axial tensile load
b) axial compressive load

can be applied to the cylinder before yielding commences according to the maximum shear stress criterion.

The yield stress of the material in simple tension is 240MN/m^2. Assume that the radial stress in the wall of the cylinder is zero.

Sketch the plane yield locus for the maximum shear stress criterion and show the two points representing the cases above.

Question 4

A THIN cylinder of internal diameter 0.3m is subjected to an internal pressure of 4MN/m^2 and to an axial torque which induces a shear stress, $\tau = (0.2/t)$ MN/m^2, where t is the thickness of the cylinder. If the maximum safe stress for the material in simple tension is 150MN/m^2 and Poisson's Ratio, $\upsilon = 0.3$, find the required thickness, t according to the TRESCA failure criteria.

Question 5

A THIN cylindrical shell 2.4m diameter is composed of plates 12mm thick. The yield stress for the material is 300MN/m^2. Calculate the internal pressure, which would cause failure according to the Maximum Shear Stress criteria.